Coefficient of Quartile Deviation:

The relative measure corresponding to QD is coefficient of QD and is defined as:

$$\text{Coefficient of Quartile Deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Merits:

- It is not affected by the extreme (highest and lowest) values in the data set.
- It is an appropriate measure of variation for a data set summarized in open-ended class intervals.
- It is a positional measure of variation; therefore it is useful in the cases of erratic or highly skewed distributions.

Limitations:

- The QD is based on the middle 50 per cent observed values only and is not based on all the observations in the data set, therefore it cannot be considered as a good measure of variation.
- It is not suitable for mathematical treatment.
- It is affected by sampling fluctuations.
- The QD is a positional measure and has no relationship with any average in the data set.

6.3.3 Mean Deviation

The Mean Deviation (MD) is defined as the arithmetic mean of the absolute deviations of the individual values from a measure of central tendency of the data set. It is also known as the average deviation.

The measure of central tendency is either mean or median. If the measure of central tendency is mean (or median), then we get the mean deviation about the mean (or median).

$$\text{MD (about mean)} = \frac{\sum |D|}{n} \qquad D = (x - \bar{x})$$

$$\text{MD (about median)} = \frac{\sum |Dm|}{n} = D_m = x - \text{Median}$$

The coefficient of mean deviation (CMD) is the relative measure of dispersion corresponding to mean deviation and it is given by

$$\text{Coefficient of Mean Deviation (CMD)} = \frac{MD\,(mean\ or\ median)}{mean\ or\ median}$$

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